

RESULTS AND EVALUATION REPORT

AI-BASED FAKE PROFILE DETECTION IN SOCIAL NETWORKS

1. INTRODUCTION

This report documents the experimental results and evaluation framework for the AI-Based Fake Profile Detection system. The purpose of evaluation is to determine how effectively different machine-learning algorithms distinguish fake profiles from genuine profiles.

Important: numerical performance values must be populated from the actual dataset and trained models. This report does not invent experimental results.

2. EXPERIMENTAL SETUP

The experiment should use a labeled dataset containing fake and genuine profiles. The data should be divided into training and testing subsets. A validation set or stratified cross-validation should be used during model selection.

Recommended environment:

- Python 3.x
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Jupyter Notebook or VS Code
- Joblib for model persistence

3. DATASET RESULTS

Record the following after preprocessing:

Total records: _____

Genuine profiles: _____

Fake profiles: _____

Number of features: _____

Training records: _____

Testing records: _____

Missing values before preprocessing: _____

Missing values after preprocessing: _____

4. CLASS DISTRIBUTION ANALYSIS

The class distribution should be visualized using a bar chart or pie chart. If the dataset is imbalanced, accuracy should not be used as the only evaluation measure. Stratification, class weights, resampling, or threshold adjustment may be considered.

5. FEATURE ANALYSIS

Important features may include account age, followers, following, follower/following ratio, post count, activity rate, profile completeness, verification status, and behavioral activity.

The analysis should identify whether individual features or combinations of features show different distributions between fake and genuine profiles.

6. MODELS EVALUATED

The recommended models are:

1. Logistic Regression – baseline.
2. Decision Tree – interpretable nonlinear model.
3. Random Forest – ensemble classifier.
4. Gradient Boosting – sequential ensemble model.
5. XGBoost – optional advanced gradient-boosting model.

7. MODEL COMPARISON TABLE

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
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Logistic Regression	_____	_____	_____	_____	_____
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Decision Tree	_____	_____	_____	_____	_____
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Random Forest	_____	_____	_____	_____	_____
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Gradient Boosting	_____	_____	_____	_____	_____
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XGBoost	_____	_____	_____	_____	_____
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8. CONFUSION MATRIX

For binary classification:

True Positive (TP): fake profile correctly classified as fake.

True Negative (TN): genuine profile correctly classified as genuine.

False Positive (FP): genuine profile incorrectly classified as fake.

False Negative (FN): fake profile incorrectly classified as genuine.

Confusion matrix:

Predicted Genuine Predicted Fake

Actual Genuine TN = _____ FP = _____

Actual Fake FN = _____ TP = _____

9. EVALUATION METRICS

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1-Score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

For this project, recall is important because false negatives represent suspicious accounts that the system fails to identify. Precision is also important because excessive false positives can increase the workload for human moderators.

10. ROC-AUC ANALYSIS

The ROC curve compares true-positive rate against false-positive rate across classification thresholds. ROC-AUC summarizes ranking performance. The selected model should be assessed using both ROC-AUC and the operational precision/recall trade-off.

11. PRECISION-RECALL ANALYSIS

When fake profiles are relatively rare, Precision-Recall analysis can provide useful information about performance on the positive class. The project should report the selected operating threshold and explain why it was chosen.

12. FEATURE IMPORTANCE

For tree-based models, feature importance can be reported to identify variables that contribute strongly to predictions. Permutation importance or SHAP can provide more detailed explanations.

Suggested result table:

Rank | Feature | Importance/Contribution

1 | _____ | _____

2 | _____ | _____

3 | _____ | _____

4 | _____ | _____

5 | _____ | _____

13. RISK SCORE RESULTS

The prediction probability can be converted into an operational risk level.

Low Risk: 0.00–0.39

Medium Risk: 0.40–0.69

High Risk: 0.70–1.00

These thresholds are examples and should be calibrated using validation data.

Example output:

Profile ID: _____

Fake Probability: _____

Genuine Probability: _____

Risk Level: _____

Model Version: _____

Top Factors: _____

14. ERROR ANALYSIS

The project should investigate false positives and false negatives rather than reporting only aggregate metrics.

False positives may occur when legitimate users have unusual behavior, very high activity, or incomplete profiles.

False negatives may occur when fake profiles imitate normal users or have sufficient historical activity to resemble genuine accounts.

15. MODEL SELECTION

The final model should be selected based on:

- F1-score and recall/precision balance.
- ROC-AUC or Precision-Recall performance.
- Computational requirements.
- Interpretability.
- Inference speed.
- Stability across validation folds.

A model with slightly lower accuracy may be preferable if it provides better recall, calibration, explainability, and operational reliability.

16. DASHBOARD RESULTS

The dashboard should present:

- Profile identifier or anonymized ID.
- Fake probability.
- Genuine probability.
- Risk category.
- Top contributing factors.
- Model version.
- Prediction timestamp.
- Analyst review status.

17. PERFORMANCE TABLE

Metric | Result

Average inference time: _____ ms

Training time: _____ s

Model size: _____ MB

Number of test samples: _____

Selected threshold: _____

18. SAMPLE RESULT INTERPRETATION

After the actual experiment is completed, a result can be described as follows:

“The selected model achieved a test F1-score of ____ and ROC-AUC of _____. Its precision of ____ indicates that ____% of profiles flagged as fake were fake in the test data, while recall of ____ indicates that the model detected ____% of the fake profiles in the test set. The confusion matrix contained ____ false positives and ____ false negatives.”

Replace all blanks with measured values.

19. LIMITATIONS

Results can be affected by dataset quality, labeling errors, class imbalance, sampling bias, feature leakage, distribution shift, and changes in attacker behavior. Test-set performance does not guarantee identical real-world performance.

20. REPRODUCIBILITY

The final results should record:

- Dataset name and version.
- Dataset source and license.
- Random seed.

- Train/test split.
- Feature list.
- Preprocessing steps.
- Model hyperparameters.
- Software versions.
- Evaluation script.
- Model version.

21. FINAL RESULTS SUMMARY

Dataset size: _____

Best model: _____

Accuracy: _____

Precision: _____

Recall: _____

F1-score: _____

ROC-AUC: _____

Inference time: _____

Selected risk threshold: _____

22. CONCLUSION

The results section should demonstrate not only which model performs best, but also why it is appropriate for fake-profile detection. The final evaluation should combine statistical metrics, error analysis, explainability, computational performance, and practical moderation requirements.